Safety Data Sheet

TargetMol®
A DRUG SCREENING EXPERT

Creation Date: May 12, 2024
Revision Date: May 12, 2024

According to the UN GHS revision 8

1. IDENTIFICATION

1.1 GHS Product identifier

Product name: N-Methylpyrrolidone

Catalog Number: T5851

CAS Number: 872-50-4

1.2 Other means of identification

Other names: -

1.3 Recommended use of the chemical and restrictions on use

Identified uses: no data available

1.4 Supplier's details

Company: Targetmol Chemicals Inc.

Uses advised against: 36 Washington Street,Wellesley Hills, Massachusetts 02481 USA

Tel/Fax: (781) 999-4286

1.5 Emergency phone number

Emergency phone number: 781-999-4286

Service hours: Monday to Friday, 9am-5pm (Standard timezone:UTC/GMT -5hours).

2. HAZARD IDENTIFICATION

2.1 Classification of the substance or mixture

Skin irritation, Category 2
Eye irritation, Category 2
Specific target organ toxicity – single exposure, Category 3
Reproductive toxicity, Category 1B

2.2 GHS label elements, including precautionary statements

Pictogram(s):

Signal word: Danger

Hazard statement(s):
H315 Causes skin irritation
H319 Causes serious eye irritation
H335 May cause respiratory irritation

Precautionary statement(s):

Prevention:
P264 Wash ... thoroughly after handling.
P280 Wear protective gloves/protective clothing/eye protection/face protection/hearing protection/...
P261 Avoid breathing dust/fume/gas/mist/vapours/spray.
P271 Use only outdoors or in a well-ventilated area.
P203 Obtain, read and follow all safety instructions before use.

Response:
P302+P352 IF ON SKIN: Wash with plenty of water/...
P321 Specific treatment (see ... on this label).
P332+P317 If skin irritation °Ccurs: Get medical help.

P362+P364 Take off contaminated clothing and wash it before reuse.
P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P304+P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P319 Get medical help if you feel unwell.
P318 IF exposed or concerned, get medical advice.

Storage: P403+P233 Store in a well-ventilated place. Keep container tightly closed.
P405 Store l°Cked up.

Disposal: P501 Dispose of contents/container to an appropriate treatment and disposal facility in accordance with applicable laws and regulations, and product characteristics at time of disposal.

2.3 Other hazards which do not resultin classification

no data available

3. COMPOSITION/INFORMATION ON INGREDIENTS

3.1 Substances

Chemical name	Common names and synonyms	CAS number	EC number
N-Methylpyrrolidone	-	872-50-4	212-828-1

4. FIRST-AID MEASURES

4.1 Description of necessary first-aid measures

General advice

no data available

If inhaled

Fresh air, rest. Refer for medical attention.

Following skin contact

Rinse skin with plenty of water or shower.

Following eye contact

First rinse with plenty of water for several minutes (remove contact lenses if easily possible), then refer for medical attention.

Following ingestion

Rinse mouth. Refer for medical attention .

4.2 Most important symptoms/effects, acute and delayed

Immediate first aid: Ensure that adequate decontamination has been carried out. If patient is not breathing, start artificial respiration, preferably with a demand valve resuscitator, bag-valve-mask device, or p°Cket mask, as trained. Perform CPR if necessary. Immediately flush contaminated eyes with gently flowing water. Do not induce vomiting. If vomiting °Ccurs, lean patient forward or place on the left side (head-down position, if possible) to maintain an open airway and prevent aspiration. Keep patient quiet and maintain normal body temperature. Obtain medical attention. Poisons A and B

4.3 Indication of immediate medical attention and special treatment needed, if necessary

Inhalation of hot vapors can irritate nose and throat. Ingestion causes irritation of mouth and stomach. Contact with eyes causes irritation. Repeated and prolonged skin contact produces a mild, transient irritation. (USCG, 1999)

5. FIRE-FIGHTING MEASURES

5.1 Extinguishing media

Use water spray, powder, alcohol-resistant foam, carbon dioxide.

5.2 Specific hazards arising from the chemical

Special Hazards of Combustion Products: Toxic oxides of nitrogen may be formed in fire. (USCG, 1999)

5.3 Special protective actions for fire-fighters

Use water spray, powder, alcohol-resistant foam, carbon dioxide.

6. ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

Personal protection: filter respirator for organic gases and vapours adapted to the airborne concentration of the substance. Collect leaking liquid in sealable containers. Absorb liquid in sand or inert absorbent. Store and dispose of according to l°Cal regulations. Wash away remainder with plenty of water.

6.2 Environmental precautions

Personal protection: filter respirator for organic gases and vapours adapted to the airborne concentration of the substance. Collect leaking liquid in sealable containers. Absorb liquid in sand or inert absorbent. Store and dispose of according to l°Cal regulations. Wash away remainder with plenty of water.

6.3 Methods and materials for containment and cleaning up

Spillage Disposal: Personal protection: filter respirator for organic gases and vapours adapted to the airborne concentration of the substance. Collect leaking and spilled liquid in sealable containers as far as possible. Absorb remaining liquid in sand or inert absorbent. Then store and dispose of according to l°Cal regulations.

7. HANDLING AND STORAGE

7.1 Precautions for safe handling

NO open flames. Above 86°C use a closed system and ventilation. Handling in a well ventilated place. Wear suitable protective clothing. Avoid contact with skin and eyes. Avoid formation of dust and aerosols. Use non-sparking tools. Prevent fire caused by electrostatic discharge steam.

7.2 Conditions for safe storage, including any incompatibilities

Dry. Ventilation along the floor. Separated from strong oxidants, strong acids, strong bases, copper and plastics.Small quantities can be stored in polyethylene, polypropylene, or clear glass bottles.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 Control parameters

Occupational Exposure limit values

EU-OEL: 40 mg/m3, 10 ppm as TWA; 80 mg/m3, 20 ppm as STEL; (skin).MAK: 82 mg/m3, 20 ppm; peak limitation category: II(2); skin absorption (H); pregnancy risk group: C

Biological limit values

no data available

8.2 Appropriate engineering controls

Ensure adequate ventilation. Handle in accordance with good industrial hygiene and safety practice. Set up emergency exits and the risk-elimination area.

8.3 Individual protection measures, such as personal protective equipment (PPE)

Eye/face protection

Wear safety goggles.

Skin protection

Protective gloves. Protective clothing.

Respiratory protection

Use l°Cal exhaust or ventilation.

Thermal hazards

no data available

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	Liquid.
Color	Clear, colorless.
Odour	Mild amine odor

Melting point/ freezing point	-24.2 °C. Atm. press.:1 013 hPa. Remarks:Accuracy:?±?0.1?°C.
Boilingpoint or initial boiling point and boiling range	204.3 °C. Atm. press.:1 015.8 hPa.
Flammability	Combustible. Gives off irritating or toxic fumes (or gases) in a fire.
Lower and upper explosion limit/flammability limit	no data available
Flash point	91 °C. Atm. press.:1 013 hPa.
Auto-ignition temperature	245 °C. Atm. press.:1 013 hPa.
Decomposition temperature	no data available
pH	pH = 7.7-8
Kinematic viscosity	dynamic viscosity (in mPa s) = 1.661. Temperature:25.0°C.
Solubility	DMSO:50 mg/mL (504.39 mM);
N-octanol-water partition coefficient	log Pow = -0.46. Temperature:25 °C.
Vapour pressure	0.32 hPa. Temperature:20 °C. Remarks:Extrapolated value from vapour pressure curve using the antoine equation.;0.46 hPa. Temperature:25 °C. Remarks:Extrapolated value from vapour pressure curve using the antoine equation.;2.54 hPa. Temperature:50 °C. Remarks:Extrapolated value from vapour pressure curve using the antoine equation.
Density and/ or relative density	1.03 g/cm3. Temperature:25 °C.
Relative vapour density	3.4 (vs air)
Particle characteristics	no data available

10. STABILITY AND REACTIVITY

10.1 Reactivity

Decomposes on heating and on burning. This produces toxic fumes including nitrogen oxides. It reacts violently with strong acids and strong bases. Attacks copper and its alloys.

10.2 Chemical stability

NMP shows unlimited shelf life in tightly closed containers, slight discoloration does not impair its quality.

10.3 Possibility of hazardous reactions

Combustible when exposed to heat, open flame, or powerful oxidizers.This amine is a very mild chemical base. It does tend to neutralize acids to form salts plus water. The amount of heat that is evolved per mole of amine in a neutralization is largely independent of the strength of the amine as a base. Amines may be incompatible with is°Cyanates, halogenated organics, peroxides, phenols (acidic), epoxides, anhydrides, and acid halides. Flammable gaseous hydrogen is generated by amines in combination with strong reducing agents, such as hydrides.

10.4 Conditions to avoid

no data available

10.5 Incompatible materials

Reacts with sulfur or carbon disulfide at high temperatures and pressures.

10.6 Hazardous decomposition products

Decomposes on heating and on burning. This produces toxic fumes including nitrogen oxides and carbon monoxide.

11. TOXICOLOGICAL INFORMATION

Acute toxicity

Oral: LD50 - rat (male/female) - 4 150 mg/kg bw.
Inhalation: LC50 - rat (male/female) - > 5.1 mg/L air.
Dermal: LD50 - rat (male/female) - > 5 000 mg/kg bw.

Skin corrosion/irritation

no data available

Serious eye damage/irritation

no data available

Respiratory or skin sensitization

no data available

Germ cell mutagenicity

no data available

Carcinogenicity

no data available

Reproductive toxicity

no data available

STOT-single exposure

The substance is irritating to the eyes and respiratory tract. The substance is mildly irritating to the skin. Exposure to very high concentrations could cause lowering of consciousness.

STOT-repeated exposure

Repeated or prolonged contact with skin may cause dermatitis. Animal tests show that this substance possibly causes toxic effects upon human reproduction.

Aspiration hazard

A harmful contamination of the air will not or will only very slowly be reached on evaporation of this substance at 20°C; on spraying or dispersing, however, much faster.

12. ECOLOGICAL INFORMATION

12.1 Toxicity

Toxicity to fish: LC50 - Oncorhynchus mykiss (previous name: Salmo gairdneri) - > 500 mg/L - 96 h.
Toxicity to daphnia and other aquatic invertebrates: EC0 - Daphnia magna - > 1 000 mg/L - 24 h.
Toxicity to algae: EC50 - Desmodesmus subspicatus (previous name: Scenedesmus subspicatus) - 600.5 mg/L - 72 h.
Toxicity to microorganisms: EC50 - activated sludge, industrial - > 600 mg/L - 30 min. Remarks:Respiration rate.

12.2 Persistence and degradability

AEROBIC: In a static die-away system using a sewage sludge seed and a semi-continuous activated sludge system, 1-methyl-2-pyrrolidinone displayed 95% removal after 2 weeks and an average 7 day biodegradability of 95%, respectively(1). 1-Methyl-2-pyrrolidinone at an initial concentration of 210 ppm underwent >98% removal after 24 hrs when in°Culated with a sewage sludge seed(2). 1-Methyl-2-pyrrolidinone at an initial concentration of 150 mg/L underwent 94% removal as measured by COD after a 1 day lag period in a screening study using an activated sludge seed(3). In a model flow-through biological treatment apparatus with an 18-hour retention time, 1-methyl-2-pyrrolidinone at an initial concentration of 300 mg/L underwent >98% removal using an activated sewage sludge(4). In a static screening test, 1-methyl-2-pyrrolidinone underwent >90% removal after a 3-5 day acclimation period using a sewage sludge seed (5). 1-Methyl-2-pyrrolidinone, present at 100 mg/L, reached 73% of its theoretical BOD in 4 weeks using an activated sludge in°Culum at 30 mg/L in the Japanese MITI test(6).

12.3 Bioaccumulative potential

An estimated BCF of 3 was calculated in fish for 1-methyl-2-pyrrolidone(SRC), using a log Kow of -0.38(1). According to a classification scheme(3), this BCF suggests the potential for bi°Concentration in aquatic organisms is low(SRC).

12.4 Mobility in soil

The K°C of 1-methyl-2-pyrrolidone is estimated as 5(SRC), using a log Kow of -0.38(1) and a regression-derived equation(2). According to a classification scheme(3), this estimated K°C value suggests that 1-methyl-2-pyrrolidone is expected to have very high mobility in soil. 1-Methyl-2-pyrrolidinone had Rf values of 0.74, 0.65, 0.67, and 1.0 in silt, loam, clay and sand, respectively, in laboratory soil thin layer chromatography (TLC) experiments(4) which is consistent with significant mobility in soil(SRC).

12.5 Other adverse effects

no data available

13. DISPOSAL CONSIDERATIONS

13.1 Disposal methods

Product

The material can be disposed of by removal to a licensed chemical destruction plant or by controlled incineration with flue gas scrubbing. Do not contaminate water, foodstuffs, feed or seed by storage or disposal. Do not discharge to sewer systems.

Contaminated packaging

Containers can be triply rinsed (or equivalent) and offered for recycling or reconditioning. Alternatively, the packaging can be punctured to make it unusable for other purposes and then be disposed of in a sanitary landfill. Controlled incineration with flue gas scrubbing is possible for combustible packaging materials.

14. TRANSPORT INFORMATION

14.1 UN Number

no data available

14.2 UN Proper Shipping Name

no data available

14.3 Transport hazard class(es)

no data available

14.4 Packing group, if applicable

no data available

14.5 Environmental hazards

no data available

14.6 Special precautions for user

no data available

14.7 Transport in bulk according to IMO instruments

no data available

15. REGULATORY INFORMATION

15.1 Safety, health and environmental regulations specific for the product in question

European Inventory of Existing Commercial Chemical Substances (EINECS)	Listed.
EC Inventory	Listed.
United States Toxic Substances Control Act (TSCA) Inventory	Listed.
China Catalog of Hazardous chemicals 2015	Not Listed.
New Zealand Inventory of Chemicals (NZI°C)	Listed.
Philippines Inventory of Chemicals and Chemical Substances (PICCS)	Listed.
Vietnam National Chemical Inventory	Listed.
Chinese Chemical Inventory of Existing Chemical Substances (China IECSC)	Listed.
Korea Existing Chemicals List (KECL)	Listed.

16. OTHER INFORMATION

Information on revision

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Abbreviations and acronyms

- CAS: Chemical Abstracts Service
- ADR: European Agreement concerning the International Carriage of Dangerous Goods by Road
- RID: Regulation concerning the International Carriage of Dangerous Goods by Rail
- IMDG: International Maritime Dangerous Goods
- IATA: International Air Transportation Association
- TWA: Time Weighted Average
- STEL: Short term exposure limit
- LC50: Lethal Concentration 50%
- LD50: Lethal Dose 50%
- EC50: Effective Concentration 50%

References

IPCS - The International Chemical Safety Cards (ICSC), website: http://www.ilo.org/dyn/icsc/showcard.home
HSDB - Hazardous Substances Data Bank, website: https://toxnet.nlm.nih.gov/newtoxnet/hsdb.htm
IARC - International Agency for Research on Cancer, website: http://www.iarc.fr/
eChemPortal - The Global Portal to Information on Chemical Substances by OECD, website: http://www.echemportal.org/echemportal/index?pageID=0&request_l°Cale=en
CAMEO Chemicals, website: http://came°Chemicals.noaa.gov/search/simple
ChemIDplus, website: http://chem.sis.nlm.nih.gov/chemidplus/chemidlite.jsp
ERG - Emergency Response Guidebook by U.S. Department of Transportation, website: http://www.phmsa.dot.gov/hazmat/library/erg
Germany GESTIS-database on hazard substance, website: http://www.dguv.de/ifa/gestis/gestis-stoffdatenbank/index-2.jsp
ECHA - European Chemicals Agency, website: https://echa.europa.eu/

Other Information

N-Methyl-2-pyrrolidone enhances the skin permeability for other substances.

Disclaimer: The above information is believed to be correct but does not purport to be all inclusive and shall be used only as a guide. The information in this document is based on the present state of our knowledge and is applicable to the product with regard to appropriate safety precautions. It does not represent any guarantee of the properties of the product. We as supplier shall not be held liable for any damage resulting from handling or from contact with the above product.
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